Analysis of iPSC Gene Expression (will be changed)

Final Project

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Introduction

The discovery of human induced pluripotent stem cells (iPSCs) by Takahashi and Yamanaka in 2007 (Takahashi et al., 2007) has revolutionized the regenerative medicine field. iPSCs are created by reprogramming somatic cells into an embryonic like state using the Yamanaka transciption factors: SOX2, OCT4, KLF4, and c-Myc). These cells have vast capabilities due to their ability to model human development and diseases *in vitro*, and their potential for patient specific (autologous) and "off the shelf" (allogeneic) cell therapies (Cerneckis et al., 2024).

With that being said, the characterization of iPSC lines is essential in any experimental or manufacturing process to ensure consistency in culturing. This includes analyzing the expression of pluripotency (stem cell) markers, differentiation markers, and potential chromosomal abnormalies, to access the population of the culture. This is commonly done using techniques such as reverse transcriptase or real time quantitative PCR (RT-qPCR) to analyze mRNA expression (Artyukhov et al., 2017), or using an antibody based experiment to analyze protein expression levels, such as Flow Cytometry or immunocytochemistry. When conducting these experiments, specifically RT-qPCR, reference genes, also known as housekeeping genes, are used to normalize the expression of other genes in a sample (Artyukhov et al., 2017). This allows the expression of genes to be compared across different samples, since it is presumed that the expression of the housekeeping gene is consistent across all of the samples. In stem cell samples, however, it has been found that the commonly used housekeeping genes, such as GAPDH, appear to fluctuate in iPSC samples (Panina et al., 2020), making it difficult to achieve accurate normalization data.

RNA Seq is a technique that sequences the entirety of RNA molecules in a given sample, rather than using gene specific primers or probes as in RT-qPCR and other techniques. This allows the entire transciptome to be analyzed, providing a complete picture of the expression of genes in a sample (Wang et al., 2009). Additionally, gene expression is calculated by dividing the raw counts of a gene by the total counts in a sample, eliminating the need for a housekeeping gene. This allows for an unbiased and quantitiative analysis of millions of genes in sample (Wang et al., 2009). Despite these advantages, RNA-Seq is far more time consuming and expensive than RT-qPCR and other experiments, making it non practical in a manufacturing setting. Therefore, utilizing RNA-Seq to analyze iPSC samples could provide insight into housekeeping gene expression, allowing researchers to select ideal reference genes for their analysis.

A similar study, conducted by Carcamo-Orive *et al.,* investigated the variability of gene expression across 317 iPSC lines generated from 101 individuals (Carcamo-Orive et al., 2017) and created a publically avaible data set of their raw RNA Seq data. They found that genetic variability in the iPSC lines was influenced by a donor specific gene expression pattern, specifically influenced by donor age, sex, ancestry, and reprogramming variabilies. iPSCs can also be reprogrammed from a variety of somatic cells, such as blood, skin fibroblasts, hair keratinocytes, urine cells, mesencymal stem cells, and bone marrow (Poetsch et al., 2022). While some somatic cells are easier to obtain than others, such as blood cells and fibroblasts, iPSCs from these lines are associated with higher genomic mutations due to the somatic cells' high turnover rate. Additionally, iPSCs appear to have an epigenetic memory of their tissue of origin, which has been found to influence their differentiation (Poetsch et al., 2022). All of these factors, and many others, could be contributing to the variations in gene expression in iPSC samples.

This publicly available dataset from Carcamo-Orive *et al.* was used to analyze the expression of RNA across iPSC samples, with a focus on the expression of pluripotency markers, differentiation markers, and housekeeping genes.

Methods

Data Cleaning

RNA-Seq counts per million data was acquired from Stemformatics (*Stemformatics Datasets*, n.d.), using the Carcamo-Orive (2017) dataset (Carcamo-Orive et al., 2017). Genes in this study were referred to by their Ensembl ID, so gene information was integrated from Emsembl database (*Ensembl Biomart*, n.d.). To begin to clean the data, the Ensembl gene ID in both the count per million data frame and Ensembl data frame were mutated, using the dplyr package (Wickham et al., 2023) from tidyverse (Wickham, 2023), to the name: Gene stable ID. The Ensembl gene data frame was then filtered for the genes in the count per million data set (reducing the number of genes from 86,369 to 56,693).

further cleaning of the count per million data will be added

Figure 1

A new column was then added to the Ensembl data frame called Category, which uses the Gene type column to categorize the genes into the following categories: Protein coding, long non coding RNA (lncRNA), Small RNAs, Pseudogenes, and Other. A count of the number of genes in each category was then performed by using group by Category and summarizing a count. A fraction and percentage were then calculated by adding a column to the data frame that divides the count by the sum of count (i.e. total count for all categories). This summarized data was later plotted as a pie chart, so modifications were made to the summarized data, such as defining ymax, ymin, and label position, for ease of plotting. Text labeled were also created, which include the category and the percentage calculated above. Percentages over 10% where then added to a subcategory, called large slice, so that large categories could be labeled on the slices but small categories would be labeled outside of the pie chart. The scales (Wickham, Pedersen, et al., 2025) package from ggplot2 (Wickham, Chang, et al., 2025) was used to create the pie chart, by plotting " " on the x axis, fraction on the y axis, and fill by category. Geom_bar and coord_polar were used to format the pie chart, using fraction for the size and angle of the slices. Theme_void was then used to remove the default plot elements, creating a white background. Plot.margin was then set to 0 on all sides to remove white space around the plot. Labels on large slices were added by using the subset, large slice categories, and adding a label using the label position and label text defined above. Repel labels were added to categories not in large slices, using the same label position and label text.

continue methods for following figures …

Results

Figure 2. Expression by RNA Type (similar to figure 1 but expression based rather than count based)

```
[1] TRUE
```

Figure 3. Expression of Pluripotency Genes (SOX2, POU5F1(OCT4), KLF4, MYC (c-Myc), TRA-1-81 and 1-60 (PODXL), NANOG, FUT4 (SSEA4), LIN28A, LIN28B, ESRRB, PRDM14, ZRP42 (REX1), TFCP2L1

(ectoderm: PAX6, NESTIN or SOX1 (neural projenitor/stem cell markers), KRT14 (epithelial), endoderm: SOX17, AFP (hepatic), PDX1 (pancreatic), mesoderm: CD34 (hematopoitic), CD90 (mesenchymal), NKX2.5 (cardiac progenitor))

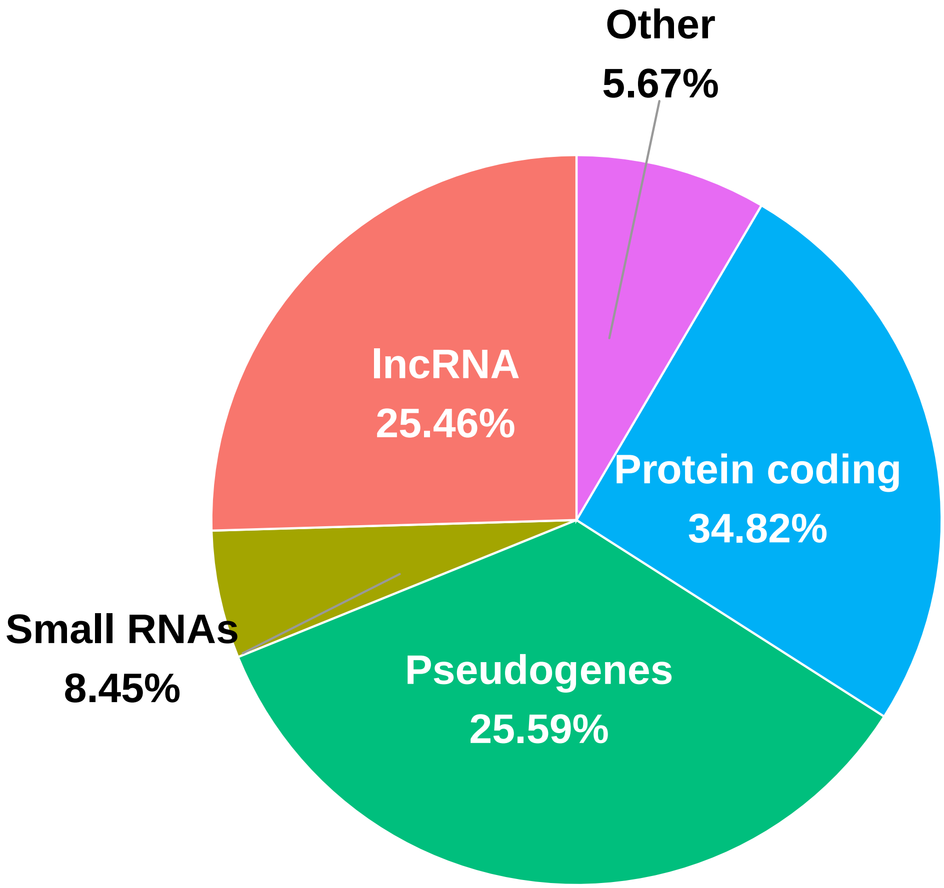

Figure 1: Detected RNA Types. The number of genes detected in each RNA category from RNA-seq data were counted and plotted as a pie chart. Small categories (< 2 %), such as rRNA, tRNA, and immunoglobulin genes, were combined into an 'Other' category.

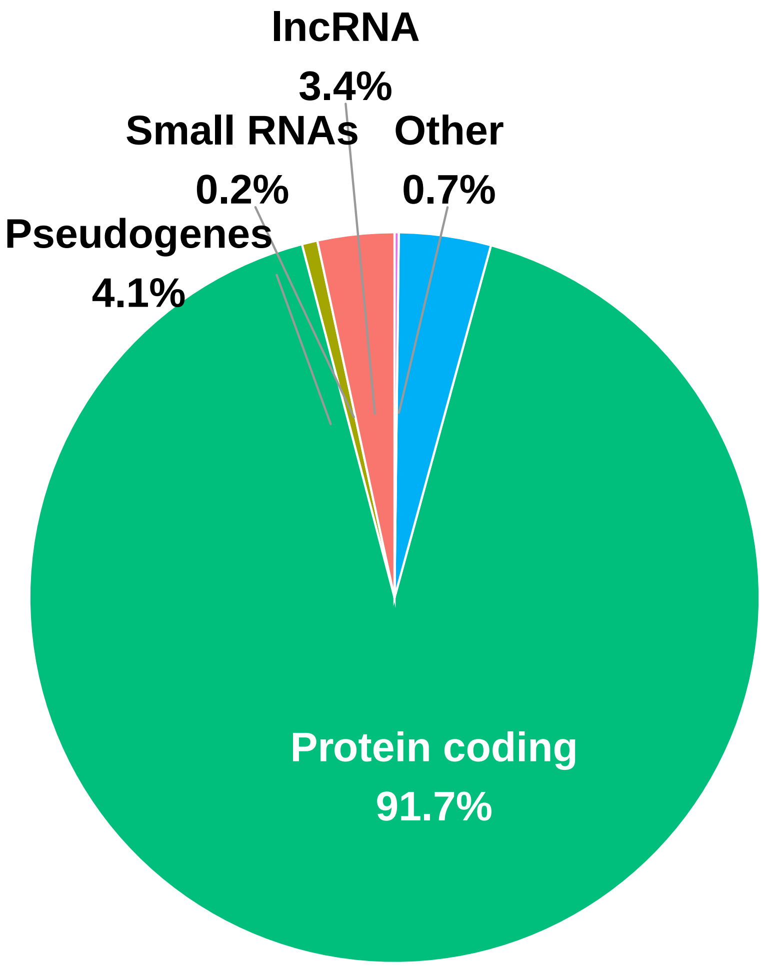

Figure 2: Expression by RNA Types. The expression of genes, using raw count per million data, was averaged by type of RNA and plotted as a pie chart. Small categories (< 2 %), such as rRNA, tRNA, and immunoglobulin genes, were combined into an 'Other' category.

```r
#create a list of pluripotency, differentiation, and housekeeping genes
pluripotency_genes <- c("SOX2", "POU5F1", "KLF4", "MYC",
                        "PODXL", "NANOG", "FUT4", "LIN28A",
                        "LIN28B", "ESRRB", "PRDM14", "TFCP2L1")

differentiation_genes <- c("SOX1","PAX6","NESTIN","KRT14",
                           "SOX17", "AFP", "PDX1", "CD34",
                           "CD90", "NKX2.5")

housekeeping_genes <- c("ACTB","GAPDH","RPL13A","HPRT1",
                        "B2M","TBP","PGK1","UBC", "RPL36AL")
```

```
character(0)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Expression ~ sex + age_decade + (1 | `Gene name`)
   Data: logCPM_pluripotency

REML criterion at convergence: 4860

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.5766 -0.5524  0.0324  0.5434  5.6678

Random effects:
 Groups    Name        Variance Std.Dev.
 Gene name (Intercept) 12.0276  3.4681
 Residual               0.2103  0.4586
Number of obs: 3696, groups:  Gene name, 12

Fixed effects:
                Estimate Std. Error         df t value Pr(>|t|)
(Intercept)      6.18060    1.00136   11.00810   6.172 6.96e-05 ***
sexmale          0.03648    0.01574 3680.00000   2.318   0.0205 *
age_decade50s   -0.02250    0.02441 3680.00000  -0.922   0.3568
age_decade60s    0.01016    0.02370 3680.00000   0.429   0.6681
age_decade70s    0.08018    0.03245 3680.00000   2.471   0.0135 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

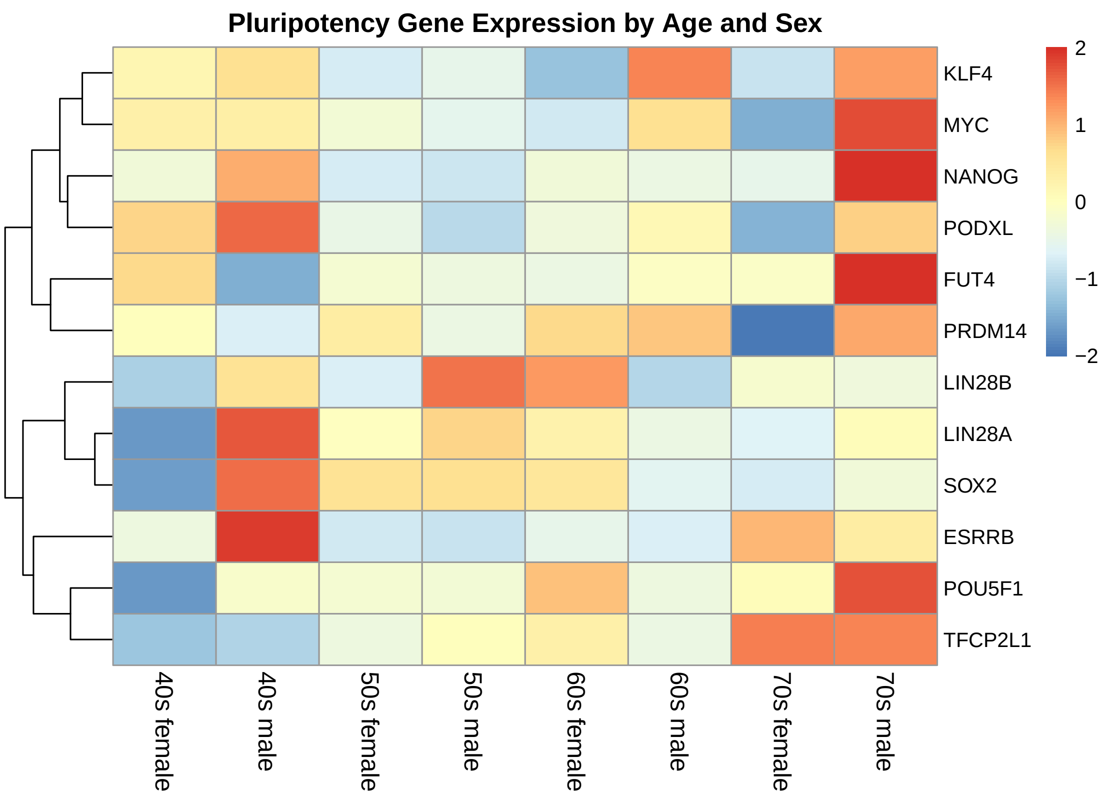

Figure 3: Expression of Pluripotency Genes

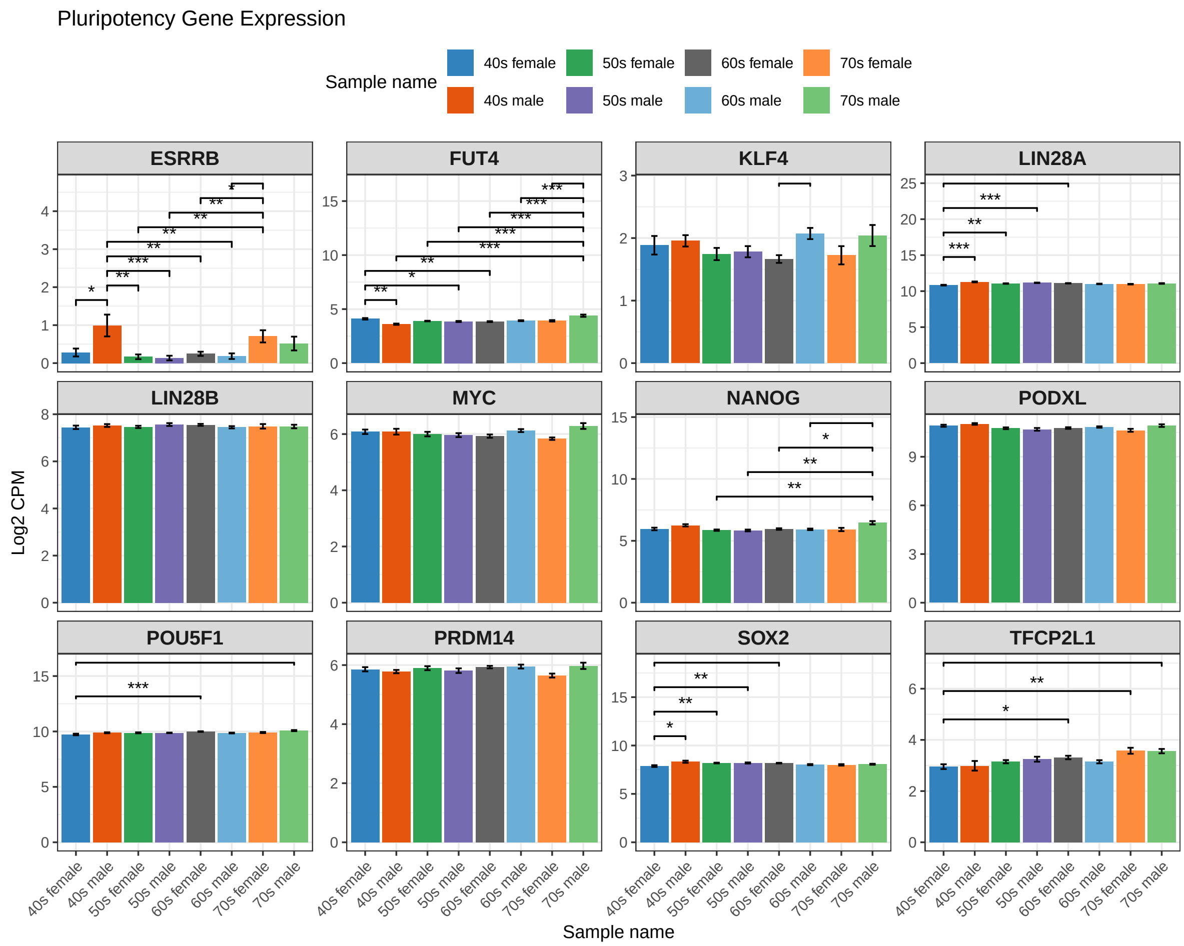

Figure 4: Expression of Pluripotency Genes

```
Correlation of Fixed Effects:
            (Intr) sexmal ag_d50 ag_d60
sexmale     -0.003
age_decd50s -0.017 -0.131
age_decd60s -0.017 -0.156  0.742
age_decd70s -0.013 -0.098  0.540  0.558

Type III Analysis of Variance Table with Satterthwaite's method
           Sum Sq Mean Sq NumDF DenDF F value   Pr(>F)
sex        1.1296 1.12964     1  3680  5.3721 0.020516 *
age_decade 2.8948 0.96492     3  3680  4.5887 0.003278 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$emmeans
 age_decade emmean SE  df asymp.LCL asymp.UCL
 40s          6.20  1 Inf      4.24      8.16
 50s          6.18  1 Inf      4.21      8.14
 60s          6.21  1 Inf      4.25      8.17
 70s          6.28  1 Inf      4.32      8.24

Results are averaged over the levels of: sex
Degrees-of-freedom method: asymptotic
Confidence level used: 0.95

$contrasts
 contrast  estimate     SE  df z.ratio p.value
 40s - 50s   0.0225 0.0244 Inf   0.922  0.7933
 40s - 60s  -0.0102 0.0237 Inf  -0.429  0.9736
 40s - 70s  -0.0802 0.0325 Inf  -2.471  0.0646
 50s - 60s  -0.0327 0.0173 Inf  -1.887  0.2335
 50s - 70s  -0.1027 0.0282 Inf  -3.643  0.0015
 60s - 70s  -0.0700 0.0275 Inf  -2.545  0.0533

Results are averaged over the levels of: sex
Degrees-of-freedom method: asymptotic
P value adjustment: tukey method for comparing a family of 4 estimates

$emmeans
 sex    emmean SE  df asymp.LCL asymp.UCL
 female   6.20  1 Inf      4.24      8.16
 male     6.23  1 Inf      4.27      8.20
```

```
Results are averaged over the levels of: age_decade
Degrees-of-freedom method: asymptotic
Confidence level used: 0.95

$contrasts
 contrast      estimate     SE  df z.ratio p.value
 female - male  -0.0365 0.0157 Inf  -2.318  0.0205

Results are averaged over the levels of: age_decade
Degrees-of-freedom method: asymptotic
```

Fixed effects Sex effect (male): Expression is slightly higher in males by 0.036 logCPM; significant at p = 0.0196.

Age effects: Only the 70s group is significantly higher than 40s (p = 0.0125). 50s and 60s are not significantly different from 40s.

Type III Analysis of Variance Table with Satterthwaite's method - confirms both sex and age_decade have a significant effect on pluripotency gene expression when averaged across all genes.

Predicted mean logCPM expression (adjusted means, not raw means)

age_decade emmean 40s 6.20 50s 6.18 60s 6.21 70s 6.28

Overall, pluripotency gene expression is slightly higher in males compared to females, regardless of age. Expression also shows an increasing trend with age, with samples from individuals in their 70s being significantly higher than those from individuals in their 40s. Differences between other age decades (50s, 60s) and 40s were not statistically significant.

```r
#ANOVA and Tukey HSD

#create empty dataframes to store results
gene_anova_results <- data.frame()
gene_tukey_results <- data.frame()

#use a for loop to loop through each gene and perform an ANOVa and tukey tests
for (g in unique(logCPM_pluripotency$`Gene name`)) {

  gene_data <- logCPM_pluripotency %>%
    filter(`Gene name` == g)

  # One-way ANOVA (Expression ~ sex_age)
  fit <- aov(Expression ~ age_sex, data = gene_data)
```

```
  # Save tidy ANOVA table
  anova_tbl <- broom::tidy(fit) %>%
    mutate(Gene = g)

  gene_anova_results <- rbind(gene_anova_results, anova_tbl)

  # Tukey HSD post-hoc for age_sex
  tukey <- TukeyHSD(fit, "age_sex")
  tukey_df <- as.data.frame(tukey$age_sex) %>%
    rename(p_value = `p adj`) %>%
    mutate(Gene = g,
           comparison = rownames(.))

  gene_tukey_results <- rbind(gene_tukey_results, tukey_df)
}

# ANOVA FDR correction
pluripotency_gene_anova_results <- gene_anova_results %>%
  group_by(term) %>%
  mutate(p_adj = p.adjust(p.value, method = "BH")) %>%
  ungroup()

# Tukey HSD p-value adjustment
pluripotency_gene_tukey_results <- gene_tukey_results %>%
  group_by(Gene) %>%
  mutate(p_adj = p.adjust(p_value, method = "BH"))

#see the number of p values less than 0.05
#sum(gene_tukey_results$p_value < 0.05, na.rm = TRUE)
```

Figure 4. Expression of Differentiation Genes (ectoderm: PAX6, NESTIN or SOX1 (neural projenitor/stem cell markers), KRT14 (epithelial), endoderm: SOX17, AFP (hepatic), PDX1 (pancreatic), mesoderm: CD34 (hematopoitic), CD90 (mesenchymal), NKX2.5 (cardiac progenitor))

Figure 5. Expression of Housekeeping Genes (some sort of general figure or sorted by housekeeping type (i.e. ribosomal, metabolic, cytoskeleton, etc.))

Figure 6. Expression of Housekeeping Genes (focusing on some genes, either over time during reprogramming or across different conditions)

these figures may be split into more than the six described depending on the data

figures will include heat maps, bar charts, and possibly volcano plots

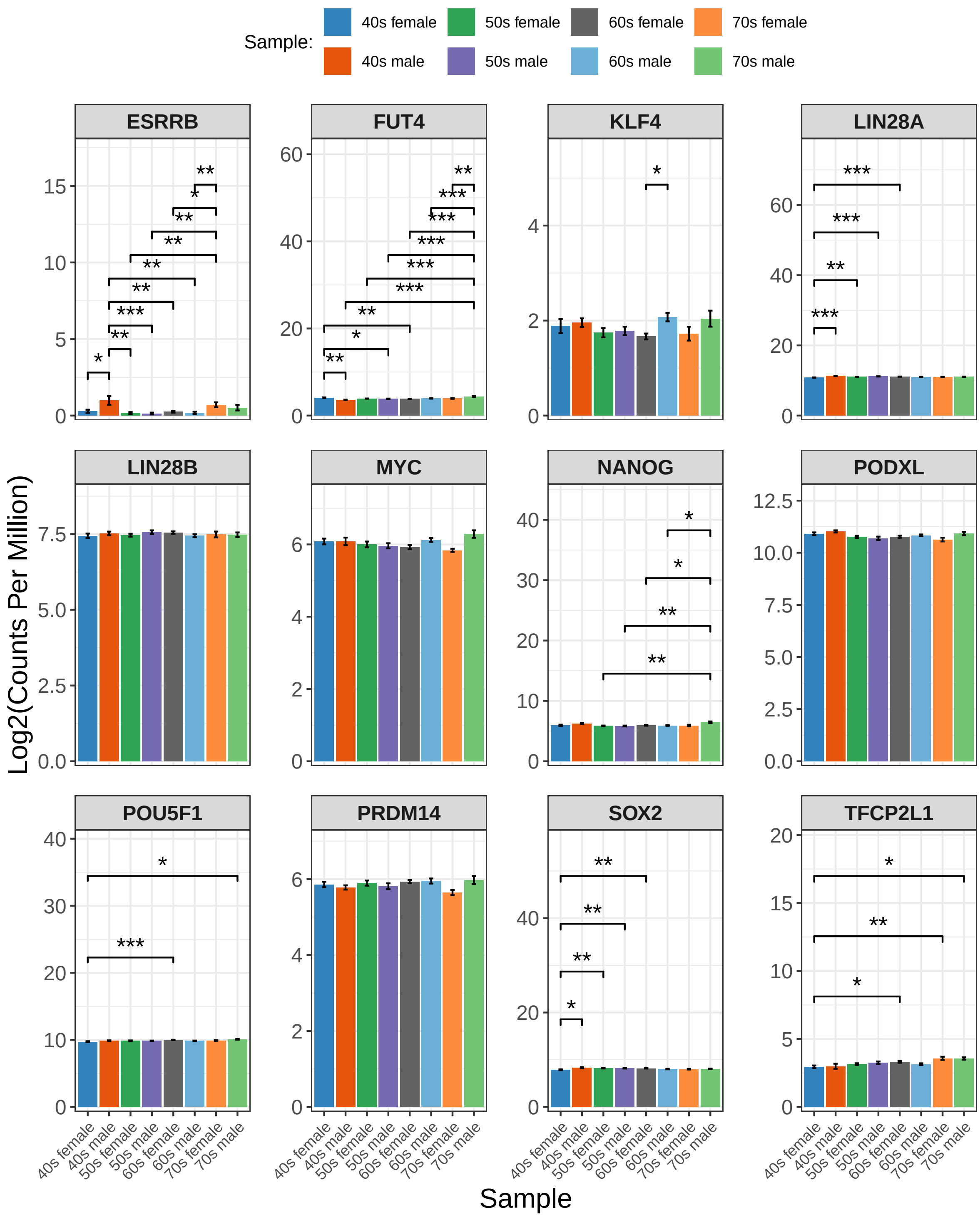

Figure 5: Pluripotency Gene Expression by Age and Sex.

Discussion

Conclusion

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